DATA ANALYTICS PROJECT REPORT

Telecom Customer Churn Analysis

Leveraging Data Analytics to Improve Customer Retention and Business Performance

1,200	24.3%	$20,179	$242,152
CUSTOMERS ANALYSED	CHURN RATE	MONTHLY REVENUE AT RISK	ANNUALISED REVENUE LOST

Dataset	telecom_customer_churn_dataset: 1,200 customers, 8 fields (demographics, tenure, contract, payment method, internet service, monthly charges, churn flag)
Deliverables	Live Excel dashboard workbook (telecom_customer_churn_dashboard.xlsx) and this report
Tools	Microsoft Excel: Excel tables, formula-driven analysis, data validation, conditional formatting, charts
Date	November 2026

Abstract

Customer churn, the rate at which customers discontinue a service, directly affects revenue, profitability and long-term growth in the highly competitive telecommunications sector. This project analyses a dataset of 1,200 telecom customers to measure churn, quantify the revenue it puts at risk, and test which customer characteristics are associated with leaving.

The analysis is delivered as a live Microsoft Excel dashboard in which every figure is calculated by formula from a single customer table. Users can filter by contract type, internet service, payment method and gender, add new customer records, and adjust thresholds, and all KPIs, charts and insights update automatically. The study finds an overall churn rate of 24.3% (291 of 1,200 customers), representing $20,179 of the $84,297 monthly recurring revenue (MRR), or about $242,000 a year.

Churn is spread remarkably evenly. Contract type, internet service, payment method, gender and tenure show no statistically meaningful differences. The one notable pattern is in pricing: customers paying $40–$60 a month churn at 29.1%, compared with 18.3% for the $60–$80 band. These results caution against relying on assumed churn drivers and point to targeted pricing review and richer data collection as next steps.

Table 1. Headline results (all customers)

1,200	291	24.3%	$84,297	$20,179	$70.25
TOTAL CUSTOMERS	CHURNED	CHURN RATE	MONTHLY REVENUE	MRR AT RISK	AVG MONTHLY CHARGE

Key findings

- **Churn is broad-based.** The gap between the highest- and lowest-churn segment is under 4 points for contract type (3.3), internet service (3.6), payment method (2.4) and gender (2.7), and none is statistically significant.
- **Common assumptions are not supported.** Month-to-month customers churn least (22.3%), not most. Electronic-check users (25.0%) churn no more than bank-transfer users (25.1%). Churned customers pay slightly less on average ($69.34) than retained customers ($70.54).
- **Pricing is the only notable signal.** The $40–$60 band churns at 29.1% against 18.3% for $60–$80 ($z = 2.85$). A supplementary chi-square test gives $p = 0.034$, which does not survive correction for testing six dimensions, so it is a lead rather than a conclusion.
- **Revenue risk is concentrated.** 88 high-value customers (paying $90 or more) account for 30.2% of all churned customers. Electronic check carries the largest share of revenue at risk ($7,498, 37.2%) even though its churn rate is not the highest.
- **Tenure is not protective.** Customers with 49–60 months (28.3%) and 61+ months (27.2%) churn most, while 25–36 months churn least (19.8%); the overall tenure test is not significant ($p = 0.40$).

1. Introduction

The telecommunications industry is one of the most dynamic and competitive sectors in the global economy. Customers can choose between many providers offering similar voice, internet and digital services, and can switch easily when they find better pricing, service quality or experience elsewhere.

As a result, retaining existing customers has become as important as acquiring new ones, and acquiring a customer is generally far more expensive than keeping one.

Customer churn refers to the share of customers who discontinue their relationship with a company within a given period. For subscription businesses, every lost customer represents immediate recurring revenue lost, the loss of future earnings, and the cost of replacing that customer. Data analytics allows organisations to replace intuition with evidence: to measure churn precisely, to see where it concentrates, and to quantify the revenue involved.

This project analyses a dataset of 1,200 telecom customers containing demographics, tenure, contract type, internet service, payment method, monthly charges and churn status. It applies data cleaning, descriptive analysis, KPI development, segmentation, statistical screening and dashboard design, and delivers the result as an interactive Excel workbook that stays up to date as the underlying data changes.

2. Problem Statement

Customer churn remains one of the most significant operational and financial challenges facing telecommunications companies. Despite substantial investment in marketing and customer acquisition, many organisations struggle to retain customers over the long term. High churn leads to declining revenue, higher operating costs and reduced competitiveness, and companies often cannot identify what drives dissatisfaction until customers have already left.

Traditional retention approaches rely on generalised campaigns or reactive service interventions that may not address the underlying causes of churn. Without detailed analysis of customer data, resources are allocated inefficiently, and assumptions about who churns and why go untested.

This project addresses the challenge by analysing telecom customer data to identify which factors are genuinely associated with churn, to quantify the revenue at risk, and to provide a live dashboard that supports proactive, evidence-based retention decisions.

3. Aim of the Study

The primary aim of this project is to analyse customer behaviour within a telecommunications company to identify the factors associated with churn, to test whether the differences observed are statistically meaningful, and to provide actionable insights that improve customer retention, customer satisfaction and business performance.

4. Project Objectives

The specific objectives of this study are:

1. To analyse the overall customer churn rate and the monthly revenue it puts at risk.
2. To identify the major factors influencing churn and test whether segment differences are statistically meaningful.
3. To evaluate the impact of contract type on customer retention.
4. To assess the relationship between monthly charges and churn.
5. To analyse customer tenure and loyalty patterns.
6. To examine the effect of payment method, internet service and gender on retention.

7. To deliver a live, filterable Excel dashboard that keeps these insights current as data changes.

5. Data and Methodology

5.1 Dataset

The dataset contains 1,200 customer records with unique customer IDs and no missing values. Tenure ranges from 1 to 71 months and monthly charges from $20.01 to $119.97. Currency is assumed to be US dollars because the source does not state one. The churn flag is 1 for customers who left and 0 for customers who stayed.

Table 2. Dataset fields

Field	Type	Description
CustomerID	Text	Unique customer identifier
Gender	Category (2)	Female, Male
TenureMonths	Whole number	Months as a customer (1 to 71)
ContractType	Category (3)	Month-to-Month, One Year, Two Year
PaymentMethod	Category (3)	Bank Transfer, Credit Card, Electronic Check
InternetService	Category (3)	DSL, Fiber Optic, None
MonthlyCharges	Currency	Monthly bill ($20.01 to $119.97)
Churn	Flag (0/1)	1 = churned, 0 = retained

5.2 Workbook architecture

The project is built as a four-sheet Excel workbook. Nothing on the dashboard is typed in or pasted: every number is a formula that reads the customer table, so changing the data changes the dashboard.

Table 3. Workbook structure

Sheet	Role
Dashboard	Executive view: four dropdown filters, 12 KPI cards, automated insights, seven charts and a contract by internet-service heatmap.
Analysis	Calculation engine: filter state, dropdown lists, editable thresholds, KPI table, six segment tables, heatmap grids, significance tests and data-quality checks.
Data	Customer-level records held as an Excel table with dropdown and range validation; new rows can be added beneath the last record (capacity 5,000).
Guide	How the workbook works, metric definitions, colour code, assumptions and limitations.

5.3 Interactivity and data integrity

- **Filters.** Four dropdowns (contract type, internet service, payment method, gender) drive every KPI card. Each segment chart still compares all values of its own dimension while respecting the other three filters, so comparisons stay visible.
- **Editable assumptions.** Thresholds for early-life tenure (12 months), high-value charge ($90), minimum sample size (100) and significance (z = 2.58) are input cells, as are the tenure and charge band limits.
- **Warnings.** The dashboard flags small samples and displays a data-check status that turns to a warning if any record has an unrecognised category, an invalid churn flag or a negative value.

- **Validation on entry.** Dropdown and range validation on the Data sheet prevents invalid categories, churn flags, tenures and charges from being entered.

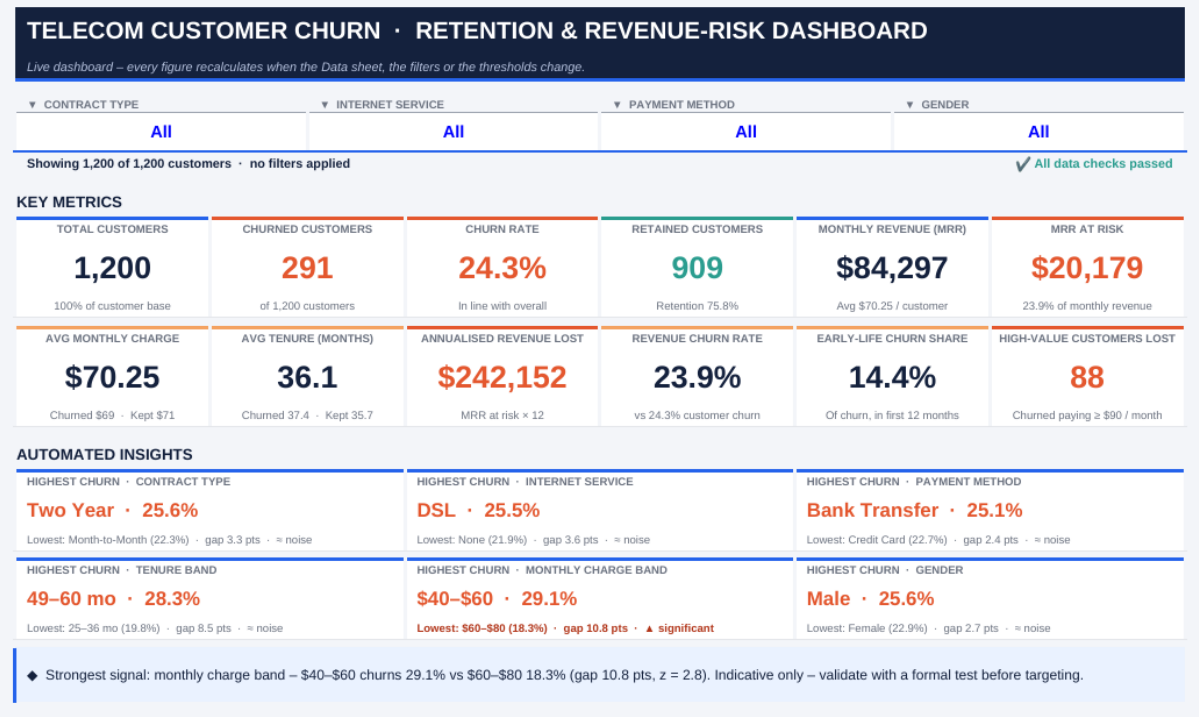


Figure 1. The live Excel dashboard: filters, KPI cards, automated insights and analyst note. (Rendered snapshot of the workbook with no filters applied.)

5.4 Metric definitions

Table 4. Key metrics

Metric	Definition
Churn rate	Churned customers divided by total customers in view.
MRR	Monthly recurring revenue: the sum of monthly charges of customers in view.
MRR at risk	MRR contributed by churned customers. Annualised revenue lost is MRR at risk multiplied by 12.
Revenue churn rate	MRR at risk divided by MRR. A value above the customer churn rate means higher-paying customers leave more often.
Early-life churn share	Share of churned customers whose tenure is 12 months or less.
High-value customers lost	Churned customers paying $90 or more per month.
Gap and z-score	Gap is the highest minus the lowest segment churn rate within a dimension. The z-score is a two-proportion test of that gap using a pooled standard error.

5.5 Statistical screening

For each dimension the workbook compares its highest-churn and lowest-churn segments with a two-proportion z-test and labels the gap Significant when |z| is at least 2.58 (about 99% confidence) and Noise otherwise. Because this compares the extreme segments after looking at the data, it is a screening tool rather than a formal hypothesis test. As supplementary validation outside the workbook, omnibus chi-square tests across all segments of each dimension were also run in Python; they are reported where relevant, together with the point that six dimensions were tested, so a Bonferroni-adjusted threshold would be about $p < 0.008$.

6. Findings

6.1 Overall churn and revenue at risk

Of 1,200 customers, 291 churned, an overall churn rate of 24.3% and a retention rate of 75.8%. Churned customers represented $20,179 of the $84,297 in monthly revenue, a revenue churn rate of 23.9%, or $242,152 per year if the loss persists. Revenue churn is marginally below customer churn because churned customers pay slightly less on average ($69.34) than retained customers ($70.54).



Figure 2. Share of customers lost (left) and share of monthly revenue lost (right).

6.2 Major factors influencing churn

Table 5 and Figure 3 screen every dimension for its widest gap. Only the monthly-charge band reaches significance. Contract type, internet service, payment method and gender differ by less than four points, and the tenure gap (8.5 points, z = 2.05) falls below the threshold. In the data as a whole, tenure and monthly charges are essentially uncorrelated with churn (r = 0.04 and r = −0.02).

Table 5. Driver screening: highest versus lowest churn segment

Dimension	Highest churn	Lowest churn	Gap (pts)	z	Signal
Monthly charge band	$40–$60 (29.1%)	$60–$80 (18.3%)	10.8	2.85	Significant
Tenure band	49–60 mo (28.3%)	25–36 mo (19.8%)	8.5	2.05	Noise
Internet service	DSL (25.5%)	None (21.9%)	3.6	1.19	Noise
Contract type	Two Year (25.6%)	Month-to-Month (22.3%)	3.3	1.09	Noise
Gender	Male (25.6%)	Female (22.9%)	2.7	1.09	Noise
Payment method	Bank Transfer (25.1%)	Credit Card (22.7%)	2.4	0.79	Noise

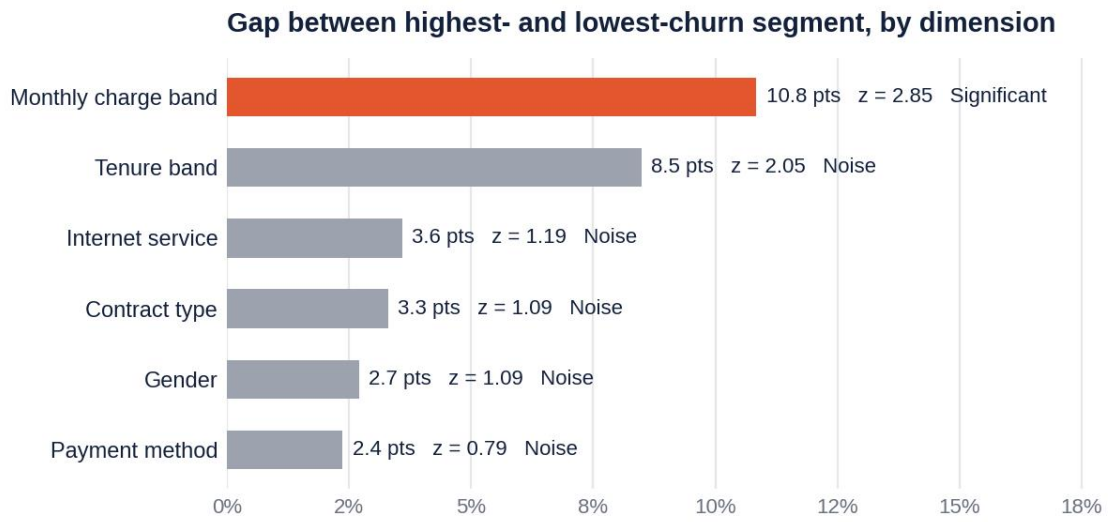


Figure 3. Gap between the highest- and lowest-churn segment, by dimension.

6.3 Contract type

Contrary to the common expectation that short contracts drive churn, month-to-month customers churned least in this dataset (22.3%, 85 of 381), followed by one-year customers (24.7%) and two-year customers (25.6%). The 3.3-point gap is not statistically significant (z = 1.09; chi-square p = 0.54). The data therefore give no evidence that promoting longer contracts would, on its own, reduce churn. Revenue at risk is highest for one-year contracts ($7,189) and two-year contracts ($7,030), and lowest for month-to-month ($5,961).

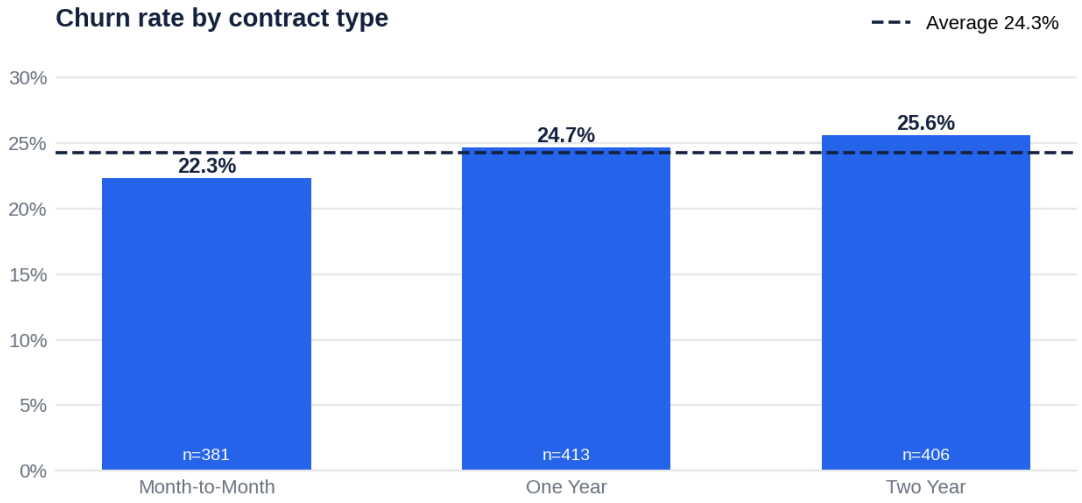


Figure 4. Churn rate by contract type. The dashed line is the average churn rate.

6.4 Monthly charges

There is no simple relationship between price and churn: churned customers pay slightly less on average than retained ones ($69.34 versus $70.54), and the correlation between monthly charge and churn is close to zero. Banding customers by price shows a non-linear pattern. Churn is highest in the $40–$60 band (29.1%, 74 of 254) and lowest in the $60–$80 band (18.3%, 45 of 246); the lowest-price and highest-price bands sit near the average (26.7% and 26.1%).

This is the only gap flagged as significant (10.8 points, z = 2.85), and the omnibus chi-square test across the five bands gives p = 0.034. Because six dimensions were examined, that result should be treated as a lead to investigate, for example whether the $40–$60 plans differ in features, promotions or

customer mix, rather than as proof of a pricing problem. High-value customers matter commercially: 88 churned customers paid $90 or more, which is 30.2% of all churn.

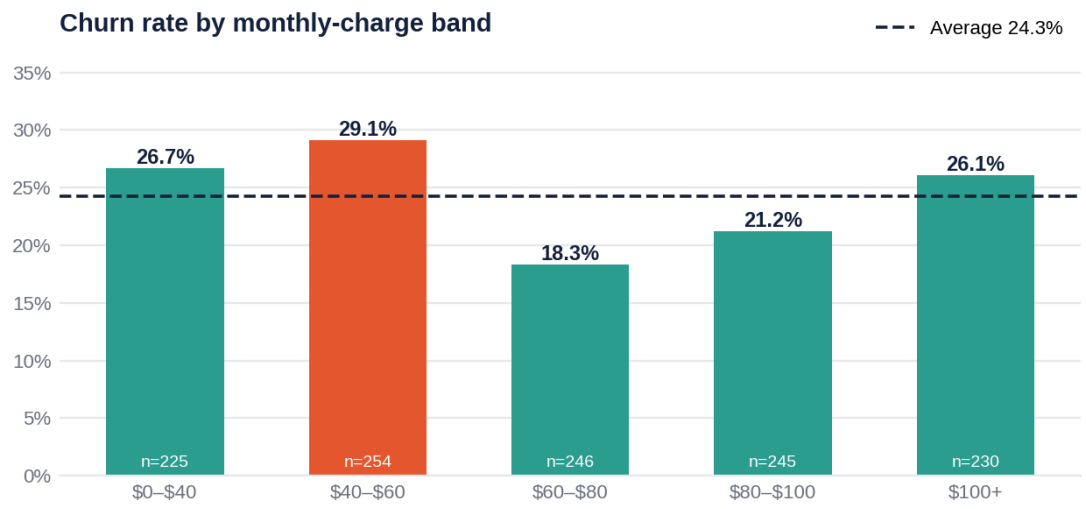


Figure 5. Churn rate by monthly-charge band. The highlighted band is the only statistically flagged gap.

6.5 Tenure and loyalty

Short tenure is not the high-risk period in this dataset. Only 42 of the 291 churned customers (14.4%) left within their first 12 months, and churned customers have a slightly longer average tenure than retained customers (37.4 versus 35.7 months). Churn is lowest at 25–36 months (19.8%) and highest after four years (28.3% at 49–60 months and 27.2% at 61 or more). The omnibus test across the six bands is not significant (p = 0.40), so this pattern could reflect chance; it does, however, argue against assuming that long-standing customers are safe.

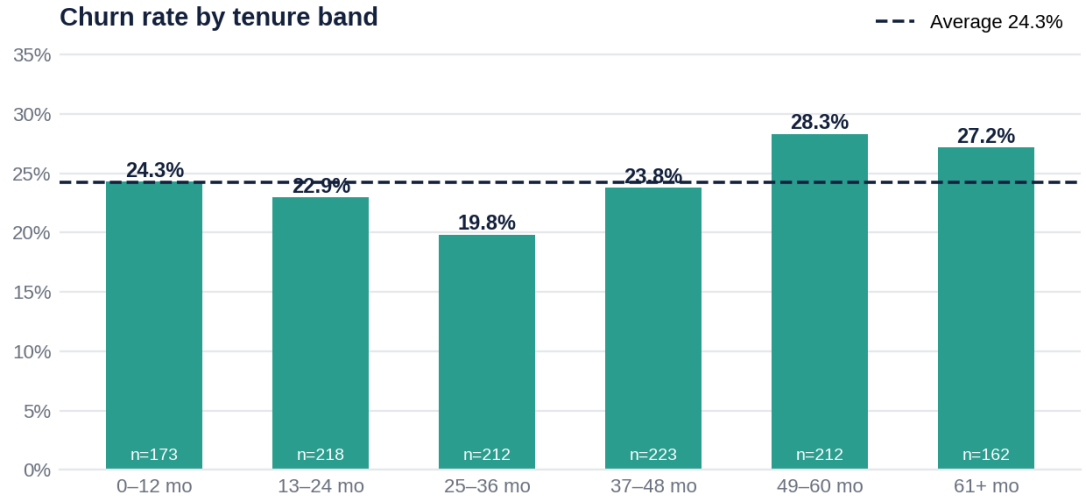


Figure 6. Churn rate by tenure band.

6.6 Payment method

Churn rates are almost identical across payment methods: bank transfer 25.1%, electronic check 25.0% and credit card 22.7% (z = 0.79; chi-square p = 0.67). Electronic-check users do not churn more than others, and the data show no consistent advantage for automatic payment methods, since bank transfer is as high as electronic check. In revenue terms, however, electronic check carries the largest exposure ($7,498, 37.2% of revenue at risk) because its customers pay more on average ($71.80 per month, compared with $68.02 for bank transfer).

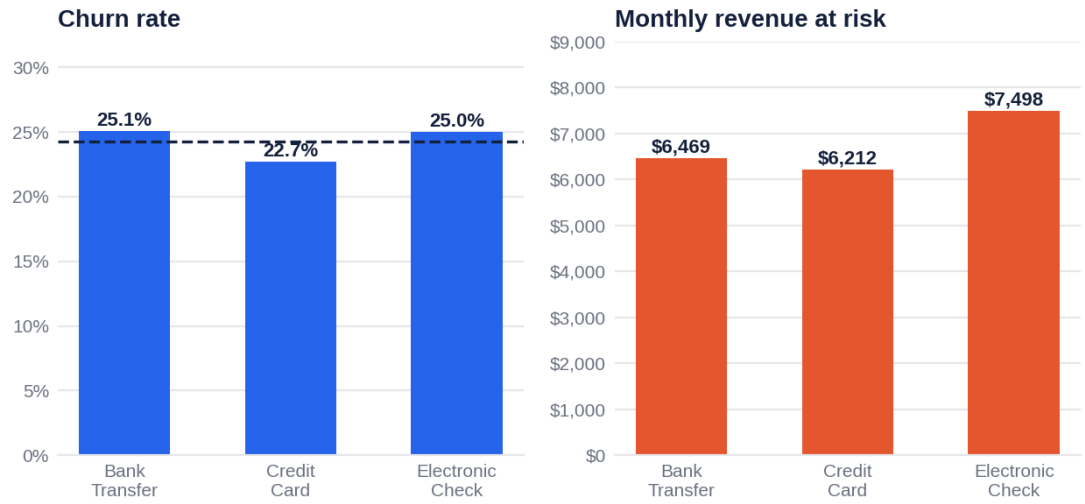


Figure 7. Churn rate (left) and monthly revenue at risk (right) by payment method.

6.7 Internet service, gender and combined segments

DSL (25.5%) and fiber optic (25.4%) customers churn at similar rates, slightly above customers without internet service (21.9%); the fiber-optic premium-service concern is therefore not visible here. Male customers churn at 25.6% versus 22.9% for female customers, a difference that is within sampling noise (z = 1.09).

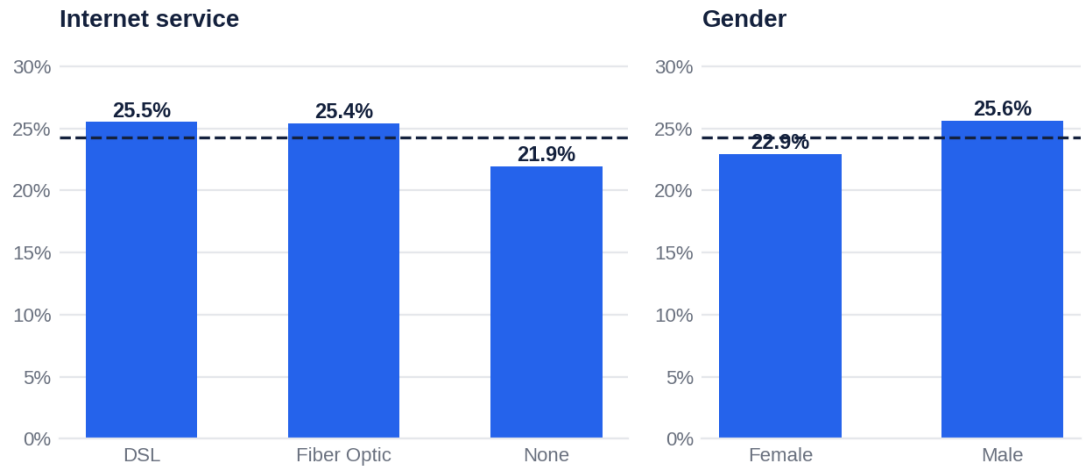


Figure 8. Churn rate by internet service (left) and gender (right).

Combining contract type and internet service gives nine segments of 118 to 160 customers each. Churn ranges from 21.0% for Month-to-Month DSL customers (n = 124) to 29.7% for One Year DSL

customers (n = 118). With samples this small, a difference of several points can arise by chance, so the heatmap is best used to spot candidates for follow-up rather than as a basis for action.

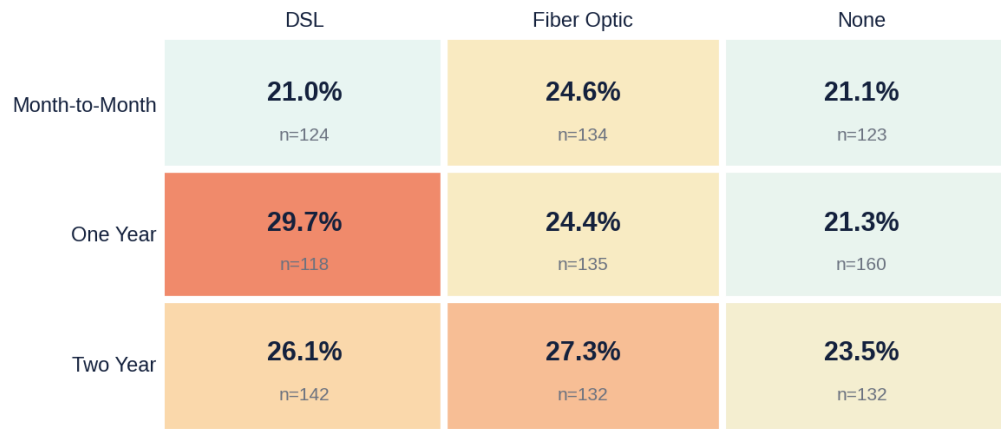


Figure 9. Churn rate by contract type and internet service, with the number of customers behind each cell.

7. Recommendations

1. **Review pricing and packaging in the $40–$60 band.** Compare plan features, promotions and customer mix with the $60–$80 band, then pilot targeted offers against a control group before any broad price change.
2. **Test assumptions before spending.** This dataset does not support campaigns built on contract conversion or payment-method switching. Run small controlled experiments first and measure the effect on churn.
3. **Protect long-tenure customers.** Customers beyond 48 months churn at 27% to 28%. Introduce renewal check-ins and loyalty recognition, and confirm whether the pattern persists in new data.
4. **Prioritise high-value accounts.** Use the dashboard threshold to monitor customers paying $90 or more, who account for 30.2% of churn, and route them to proactive service.
5. **Enrich the data.** Add churn reason codes, service start dates, usage, support contacts, complaints and competitor offers. Current fields explain very little of who leaves, so a predictive model (for example logistic regression with a hold-out sample) would only be worthwhile with richer inputs.
6. **Operationalise the dashboard.** Append new customer records monthly, review the data-check status and sample-size warnings, and track churn rate, revenue churn and the strongest-signal insight over time.

8. Significance of the Study

This study shows how data analytics can address a real business problem in the telecommunications industry. It also shows the value of testing common beliefs against data: in this dataset, several widely held churn drivers do not appear, and quantifying revenue at risk shows where retention effort could matter most.

The workbook demonstrates a reusable analytical framework: a single validated data table, a transparent calculation layer, a dashboard that responds to filters and new data, and built-in warnings about data quality and sample size. The same structure can be applied to other subscription-based businesses.

From a professional perspective, the project demonstrates data cleaning and validation, exploratory analysis, KPI development, statistical screening, dashboard design and clear business reporting.

9. Expected Business Impact

Acting on the recommendations can produce the following benefits:

- Reduced customer churn and improved retention rates through targeted rather than generalised interventions.
- Protected recurring revenue, with attention focused on high-value customers and the pricing band showing elevated churn.
- Better pricing decisions grounded in measured churn by price band.
- Faster, more confident decisions from a live dashboard that updates as data arrives.
- More efficient use of retention budgets by testing assumptions before scaling them.
- A stronger data foundation for predictive modelling and personalised retention campaigns.

As an illustration, reducing churn by one percentage point would retain about 12 customers. At the average charge of $70.25 this is roughly $843 of monthly revenue, or about $10,100 a year. This is an indicative calculation, not a forecast.

10. Limitations

- **Single period, no time dimension.** The data do not show when customers left or how behaviour changed, so trends and cohorts cannot be analysed.
- **Limited variables.** The eight fields leave most reasons for churn unobserved. Segment differences are small and the sample per segment is a few hundred customers.
- **Atypical pattern.** Churn that is nearly uniform across segments, and month-to-month contracts churning least, differs from what telecom benchmarks usually show. This suggests the dataset may be simulated or sanitised, so findings should be validated on production data before operational use.
- **Post-hoc testing.** The significance screen compares extreme segments and tests six dimensions, so a single flagged result should be treated as a lead.
- **Workbook scope.** Formulas read up to 5,000 customer rows, and new category values must be added to the lists on the Analysis sheet. Currency is assumed to be US dollars.

11. Conclusion

Customer retention is a critical determinant of long-term success in telecommunications. This project measured an overall churn rate of 24.3%, equal to $20,179 of monthly revenue and about $242,152 a year, and tested which customer characteristics are associated with it.

The central finding is that churn in this dataset is broadly uniform. Contract type, payment method, internet service, gender and tenure do not separate churners from stayers in a statistically meaningful way, and several common assumptions are contradicted. Only the monthly-charge band stands out, and even that result calls for confirmation. Recognising when the data do not support a story is itself a valuable outcome, because it prevents retention budgets from being spent on the wrong levers.

The accompanying Excel dashboard turns the analysis into a living tool. Because every figure is formula-driven, the same workbook can be refreshed with new data, filtered by segment and monitored over time, and it will flag when samples are too small or the data need checking. Combined with richer data and controlled retention experiments, it provides a practical foundation for evidence-based customer retention.

Appendix A. Segment results

All figures are for the full customer base with no filters applied. MRR at risk is the monthly revenue of churned customers.

Table 6. Customers, churn and revenue by segment

Segment	Customers	Churned	Churn rate	MRR	MRR at risk
Contract type					
Month-to-Month	381	85	22.3%	$26,914	$5,961
One Year	413	102	24.7%	$29,294	$7,189
Two Year	406	104	25.6%	$28,089	$7,030
Internet service					
DSL	384	98	25.5%	$27,416	$6,996
Fiber Optic	401	102	25.4%	$28,011	$6,797
None	415	91	21.9%	$28,870	$6,387
Payment method					
Bank Transfer	399	100	25.1%	$27,139	$6,469
Credit Card	401	91	22.7%	$28,437	$6,212
Electronic Check	400	100	25.0%	$28,721	$7,498
Gender					
Female	619	142	22.9%	$44,464	$10,198
Male	581	149	25.6%	$39,833	$9,981
Tenure band					
0–12 mo	173	42	24.3%	$12,635	$2,942
13–24 mo	218	50	22.9%	$15,531	$3,694
25–36 mo	212	42	19.8%	$14,675	$2,860
37–48 mo	223	53	23.8%	$15,175	$3,300
49–60 mo	212	60	28.3%	$15,351	$4,337
61+ mo	162	44	27.2%	$10,931	$3,047
Monthly-charge band					
$0–$40	225	60	26.7%	$6,860	$1,835
$40–$60	254	74	29.1%	$12,758	$3,778
$60–$80	246	45	18.3%	$17,323	$3,184
$80–$100	245	52	21.2%	$22,070	$4,739
$100+	230	60	26.1%	$25,285	$6,643

Appendix B. Using the dashboard

1. **Filter.** Choose values in the four dropdowns at the top of the Dashboard sheet. Set a dropdown back to All to remove its filter.
2. **Add data.** Type or paste new customers directly under the last row of the Data sheet. The table expands and all figures update.
3. **Adjust assumptions.** Change the yellow input cells on the Analysis sheet (section 2 for thresholds, sections 4e and 4f for band limits).
4. **Check quality.** Confirm the status at the top right of the Dashboard reads that all data checks passed; otherwise open Analysis section 7.
5. **Interpret with care.** Heed the small-sample warning and the Significant or Noise labels on each insight card.